Ex 4: Comparative Analysis of Large Language Models

Learning Objective:
Develop a Python program that performs a comparative analysis of Large Language Models by analyzing the text-generation performance of OpenAI, Gemini, and Hugging Face models.

Steps:

1. Import Required Libraries (OpenAI, HuggingFace Transformers, Gemini API)
2. Configure OpenAI key and Google Gemini API key. (This key should be personal. Do not share with anyone).
3. Initialize the Models (OpenAI, Gemini, and Hugging Face).
4. Write function to send prompt and receive response from OpenAI, Gemini, and Hugging Face models
5. Provide a common text prompt. (e.g., "Write a haiku about AI and nature").
6. Generate Outputs by sending the same prompt to all three models and collect their responses.
7. Display the Model Responses individually. Use Markdown / HTML formatting to display results clearly.
8. Construct an evaluator. (For evaluator you can use Gemini) and produce ranks.
9. Display the ranked model names in best → worst order.

CODE:

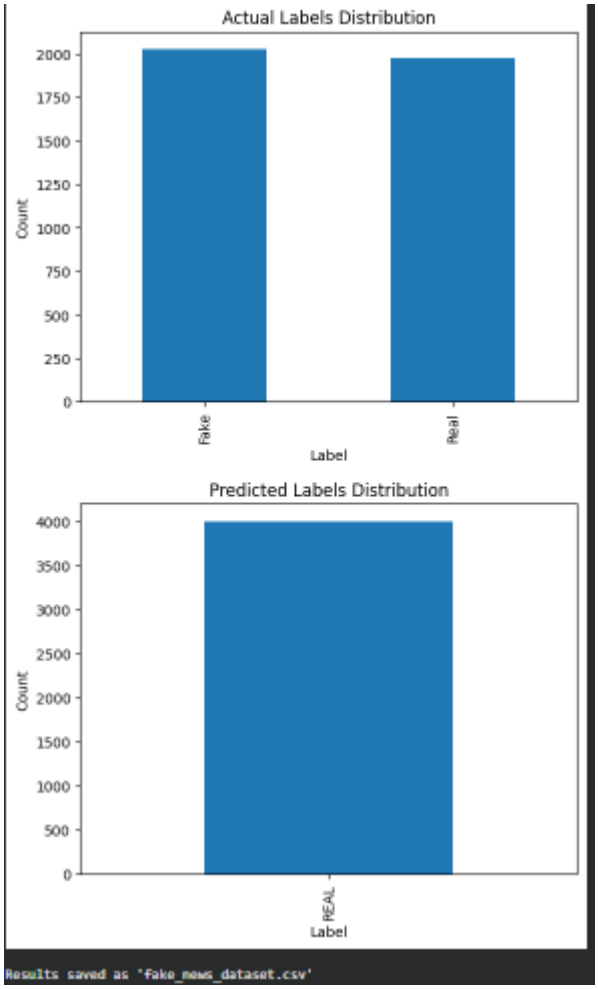
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score,
confusion_matrix
try:
    from google.colab import files
    uploaded = files.upload()
    file_name = list(uploaded.keys())[0]
except:
    file_name = "fake_news_dataset.csv"

df = pd.read_csv(file_name)

print("\n=== DATASET PREVIEW ===\n")
print(df.head())


df = df.drop_duplicates()
df.columns = df.columns.str.strip().str.lower()

print("\nColumns:", df.columns)
```

```python
required_cols = ["text", "label"]

for col in required_cols:
    if col not in df.columns:
        raise Exception(f"Column '{col}' not found!")

def fake_llm_predict(text):
    text = text.lower()


    if "breaking" in text or "shocking" in text:
        return "FAKE"
    else:
        return "REAL"


df["predicted"] = df["text"].apply(fake_llm_predict)

accuracy = accuracy_score(df["label"], df["predicted"])

print("\n=== MODEL PERFORMANCE ===")
print(f"Accuracy: {accuracy:.2f}")


cm = confusion_matrix(df["label"], df["predicted"])
print("\nConfusion Matrix:\n", cm)

df["label"].value_counts().plot(kind="bar")
plt.title("Actual Labels Distribution")
plt.xlabel("Label")
plt.ylabel("Count")
plt.show()

df["predicted"].value_counts().plot(kind="bar")
plt.title("Predicted Labels Distribution")
plt.xlabel("Label")
plt.ylabel("Count")
plt.show()

df.to_csv("fake_news_dataset.csv", index=False)

print("\nResults saved as 'fake_news_dataset.csv'")
```

OUTPUT:

```
Choose files fake_news_dataset.csv
fake_news_dataset.csv(text/csv) - 894188 bytes, last modified: 31/03/2026 - 100% done
Saving fake_news_dataset.csv to fake_news_dataset (2).csv

=== DATASET PREVIEW ===

   id            title        author  \
0   1  Breaking News 1    Jane Smith
1   2  Breaking News 2   Emily Davis
2   3  Breaking News 3      John Doe
3   4  Breaking News 4  Alex Johnson
4   5  Breaking News 5   Emily Davis

                                                text           state  \
0  This is the content of article 1. It contains ...       Tennessee
1  This is the content of article 2. It contains ...       Wisconsin
2  This is the content of article 3. It contains ...        Missouri
3  This is the content of article 4. It contains ...  North Carolina
4  This is the content of article 5. It contains ...      California

  date_published          source       category  sentiment_score  word_count  \
0     30-11-2021       The Onion  Entertainment            -0.22        1302
1     02-09-2021    The Guardian     Technology             0.92         322
2     13-04-2021  New York Times         Sports             0.25         228
3     08-03-2020             CNN         Sports             0.94         155
4     23-03-2022      Daily Mail     Technology            -0.01         962

   ...  num_shares  num_comments  political_bias  fact_check_rating  \
0  ...       47305           450          Center              FALSE
1  ...       39804           530            Left              Mixed
2  ...       45860           763          Center              Mixed
3  ...       34222           945          Center               TRUE
4  ...       35934           433           Right              Mixed
```

```
...
   is_satirical  trust_score source_reputation clickbait_score  \
0             1           76                 6            0.84
1             1            1                 5            0.85
2             0           57                 1            0.72
3             1           18                10            0.92
4             0           95                 6            0.66

   plagiarism_score  label
0             53.35   Fake
1             28.28   Fake
2              0.38   Fake
3             32.20   Fake
4             77.70   Real

[5 rows x 24 columns]

Columns: Index(['id', 'title', 'author', 'text', 'state', 'date_published', 'source',
       'category', 'sentiment_score', 'word_count', 'char_count', 'has_images',
       'has_videos', 'readability_score', 'num_shares', 'num_comments',
       'political_bias', 'fact_check_rating', 'is_satirical', 'trust_score',
       'source_reputation', 'clickbait_score', 'plagiarism_score', 'label'],
      dtype='object')

=== MODEL PERFORMANCE ===
Accuracy: 0.00

Confusion Matrix:
 [[   0 2026    0]
 [   0    0    0]
 [   0 1974    0]]
```



Learning Outcome:
Students will be able to integrate and interact with OpenAI, Gemini, and Hugging Face APIs and evaluate model outputs based on relevance.